NeuroGuard AI: A Governed Multi-Agent System for Stroke-Risk Clinical Decision Support

Sandip Dey · Udacity AI Mastery Capstone, Project 7 (Integrative Industry Synthesis) · Code: https://github.com/sandipdey2/neuroguard-agentic-rag · Live demo: https://neuroguard-agentic-rag-4ywylkye6e2rgzavdb5xrb.streamlit.app/

Overview and Industry Context

Stroke is a leading cause of serious long-term disability in the United States, where more than 795,000 people have a stroke every year (Centers for Disease Control and Prevention [CDC], 2024). Most of its drivers—high blood pressure, diabetes, heart disease, smoking—are measurable in routine primary care, yet clinicians have only minutes per visit to combine a risk estimate, the relevant guideline wording and a clear explanation. Generative AI could draft that synthesis, but in healthcare an unsupported sentence, an invented number or a dosing suggestion is a safety event, not a style problem.

NeuroGuard AI is my answer to that problem: a decision-support prototype that turns a structured patient profile and an optional free-text note into a cited, verified stroke-risk report for a clinician or a patient. It is deliberately not a medical device. Under the U.S. Food and Drug Administration's (2022) guidance, software that supports, rather than replaces, a professional's judgement must let that professional review the basis of each recommendation; NeuroGuard is designed around that idea. Every number is traceable to a tool output, every guideline claim cites a public-domain CDC or NHLBI excerpt, and a gate refuses to release reports that fail its checks. All data are public: the Stroke Prediction Dataset (fedesoriano, 2021; 5,110 records, 5,109 after cleaning), MTSamples transcriptions (Boyle, n.d.; CC0) and 30 verbatim guideline excerpts.

Integration Rationale

The system is built from six earlier capstone projects, each chosen because it solves one part of the clinical workflow.

Prior project	Contribution to NeuroGuard
P1 Data workflow (Titanic)	Explicit cleaning steps with a log: schema check, "N/A" BMI to missing, rare-category filter, imputation fitted only on the training split (`core/data.py`).
P2 Statistical analysis (heart failure, Welch's t-test)	Chi-square and Welch tests, odds ratios with 95% CIs and Wilson intervals; served to agents by the `cohort_stats` MCP tool.
P3 Classification (breast cancer, logistic regression)	Logistic regression, random forest and gradient boosting with class weighting; calibrated logistic model served with exact factor contributions.
P4 Sequence model (IMDB LSTM)	Text modelling for notes: TF-IDF + logistic regression on MTSamples, plus CDC BE FAST red-flag rules.
P5 Generative model (MNIST GAN)	Generation moved from pixels to language: the Clinical Writer produces the report from an MCP prompt template.
P6 Agentic RAG	Retrieval and an agent loop, now with a real MCP server, multiple agents and enforced verification.

The integration is causal, not decorative. P2's odds ratios (age 65+: 8.13, hypertension: 3.7, heart disease: 4.71) justify which factors the evidence agent searches for and give the writer cohort context; P3 supplies the number that P2's statistics put in context; P4's rules guard the front door; P6's lessons shape the governance. A

Project 6 reviewer correctly noted that I had described an in-process class as an "MCP server" and a single routing step as a loop. This project fixes both: the server is a separate process that speaks the Model Context Protocol (Anthropic, 2024) over stdio, and the orchestrator loop is a real graph cycle with a cap.

System Design and Technical Decisions

Figure 1 shows the architecture as implemented; every box names the file that implements it.

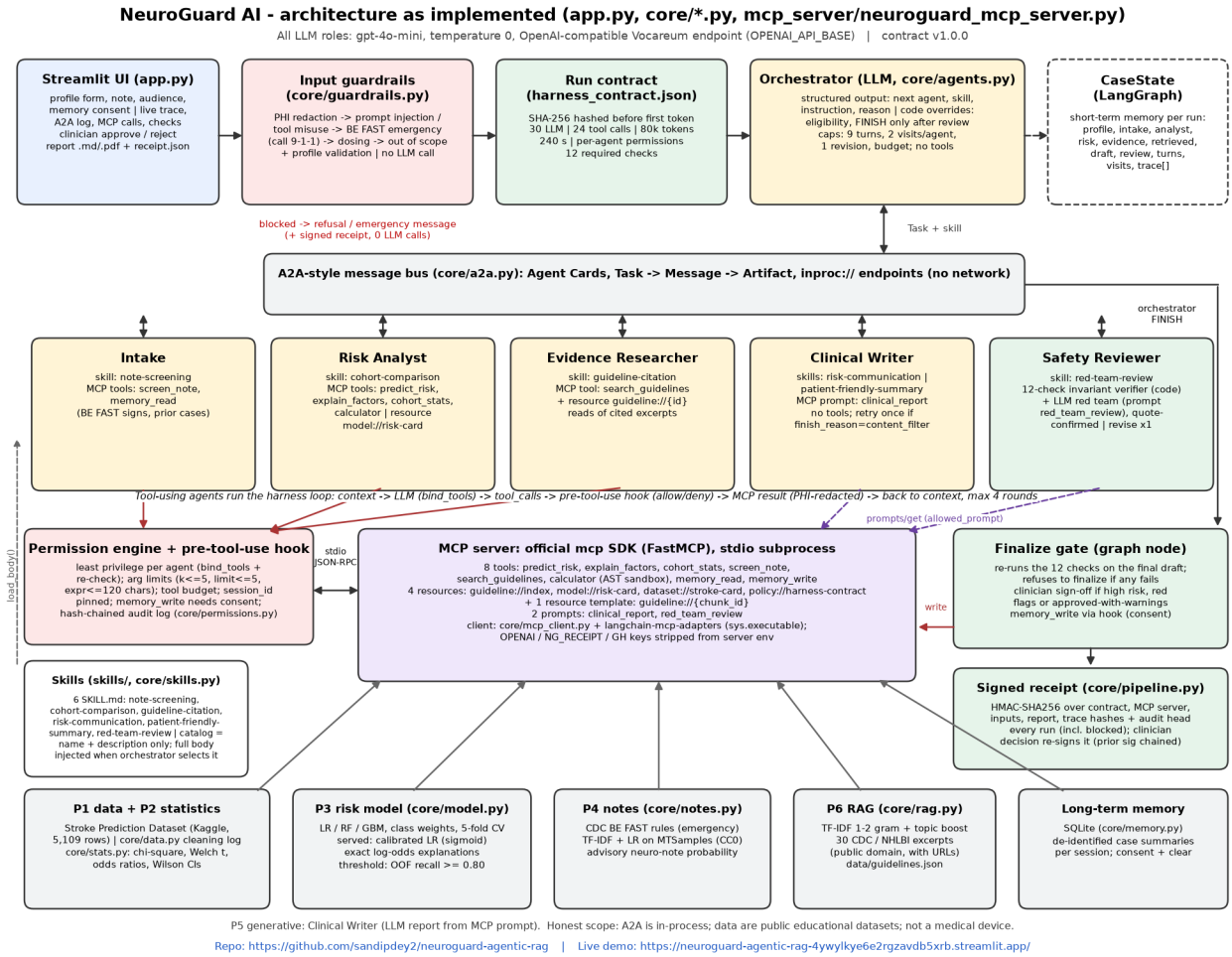


Figure 1. NeuroGuard AI architecture as implemented

Harness and contract. Before the first token, `harness_contract.json` is loaded and SHA-256 hashed. It fixes the model (gpt-4o-mini via the Vocareum OpenAI-compatible endpoint, temperature 0), nine orchestrator turns, budgets of 30 LLM calls, 24 tool calls, 80,000 tokens and 240 seconds, each agent's permitted tools, resources and prompts, and twelve required checks. The hash goes into the run receipt, so a reviewer can tell which rules governed a run.

Multi-agent loop. A LangGraph orchestrator (an LLM with structured output) chooses the next agent, a skill and an instruction; five specialists—Intake, Risk Analyst, Evidence Researcher, Clinical Writer and Safety Reviewer—return to it after every step. Code enforces eligibility: the three information-gathering agents may run in any order, the writer only after all three, the reviewer after each draft, and FINISH only after the latest draft is reviewed, with one revision allowed. Tool-using agents run a bounded ReAct loop (Yao et al., 2023): context, LLM, tool call, hook, result, back to context. Delegation uses A2A-style Agent Cards and task/message/artifact objects modelled on the Agent2Agent specification (A2A Project, 2025). Honestly scoped,

this bus is in-process rather than a networked A2A server, because a network hop would add latency without adding safety in a single deployment.

Real MCP. The official SDK server exposes eight tools (risk prediction, factor explanation, cohort statistics, note screening, guideline search, a sandboxed calculator, memory read and memory write), four resources (guideline index, model card, dataset card, contract) plus a per-excerpt resource template and two prompts (report template and red-team review). The client strips the API key from the subprocess environment.

Permissions, memory and skills. Each agent is bound only to its own tools, and a pre-tool-use hook re-checks every call, clamps arguments, pins the session identifier so no agent can read another session, enforces the tool budget and blocks `memory_write` without consent. Short-term memory is the LangGraph state; long-term memory is a consent-based SQLite table of de-identified summaries with a clear button. Skills are `SKILL.md` files: only their names and descriptions are offered to the orchestrator, and the full body is injected when one is selected (progressive disclosure).

Retrieval and modelling choices. Retrieval is TF-IDF over unigrams and bigrams with a small topic boost, following the retrieve-then-generate pattern (Lewis et al., 2020); with 30 short excerpts, sparse retrieval reached hit@3 of 1.0 on twelve hand-labelled queries, so embeddings were not worth an extra dependency. For risk, the three candidate models scored cross-validated ROC-AUC of 0.838 (logistic regression), 0.835 (random forest) and 0.822 (gradient boosting). I served the logistic model, wrapped in sigmoid calibration, because it was best and because its contributions to the log-odds are exact rather than approximated; calibration matters because class weighting inflates raw probabilities (Niculescu-Mizil & Caruana, 2005). For notes I chose TF-IDF over the P4 LSTM: it trains in seconds on Streamlit Community Cloud's CPU, needs no PyTorch wheel and its weights are inspectable.

Verifier, red team, gate and receipt. The Safety Reviewer runs deterministic invariants—risk in [0, 1], risk equal to a `predict_risk` call on the exact profile, every evidence bullet cited, citations actually retrieved, every number in the report found in a tool output or excerpt, no dosing or diagnosis language, disclaimer present, no PHI, budget respected—plus an LLM red-team pass. During development the red team, on gpt-4o-mini, flagged harmless sentences, so its findings are kept only when the quoted sentence really occurs in the draft and, for rules with a visible surface form, shows it. The finalize gate re-runs the checks and refuses to finalize on any failure; high-risk results and notes with red-flag signs require clinician sign-off. Each run ends with an HMAC-SHA256 receipt over hashes of inputs, report, trace and checks, and a hash-chained audit log.

Ethical Considerations and Responsible AI

The risks are specific to this system. First, **harm from false reassurance or delay**: a person describing sudden face drooping must not receive a risk report. Input guardrails therefore run before any LLM call, and the BE FAST rule stops the request with CDC's "call 9-1-1" wording (CDC, 2026). Second, **unsafe advice**: dosing questions are refused at input, and dosing or diagnosis language fails the verifier at output. Third, **privacy**: emails, phone numbers, record numbers and titled names are redacted before processing, memory stores only de-identified summaries with explicit consent, and Streamlit Community Cloud is documented as unsuitable for real PHI. Fourth, **bias and data provenance**: the dataset's source is described as confidential and for educational use, its binary label records whether a patient had a stroke without any timing (so estimates describe association with stroke history, not future incidence), and it contains no race or ethnicity field, so the model cannot be audited for the kind of group bias that Obermeyer et al. (2019) found in a deployed health algorithm. The app states this limitation, frames estimates as associations within an educational sample and never as population incidence, and the product zip excludes the dataset so buyers must license their own data. Fifth, **accountability and transparency**, mapped to the Govern, Map, Measure and Manage functions of the

NIST AI Risk Management Framework (National Institute of Standards and Technology [NIST], 2023) and to the WHO's (2021) call for human oversight: the contract, audit chain, signed receipt and approval step make every run reviewable by a named human.

Evaluation and Reflection

`evaluation.ipynb` was executed against the real endpoint and MCP server on eight real held-out profiles and eight attacks. The risk model's held-out results were ROC-AUC 0.84, PR-AUC 0.26 (prevalence 0.0487) and Brier score 0.041; at the recall-oriented threshold it reached recall 0.8 but precision only 0.137, which is acceptable for a prompt to discuss risk but would be unacceptable as a screening decision (Saito & Rehmsmeier, 2015). The note classifier is weak (F1 0.31), so it is advisory only.

Agent metric	**Result**
Attacks blocked (injection, tool misuse, emergency, dosing, off-topic)	1.0 (reason correct 1.0)
Normal requests wrongly blocked	0.0
Routes satisfying the policy / LLM route choices accepted without override	1.0 / 0.82
Reports finalized / first-draft pass rate	1.0 / 0.875
Citation coverage / LLM-judged faithfulness (Es et al., 2024)	1.0 / 1.0
Word-overlap support of evidence bullets (mean)	0.612
Mean latency / LLM calls / MCP tool calls per run	56.3 s / 17.6 / 8.4

The most useful finding was a failure. An earlier full run finalized only three of eight reports. The cause was not the model's reasoning: the writer pasted the long BE FAST passage verbatim and the provider stopped generation with `finish_reason="content_filter"`, truncating the citation and disclaimer. The gate refused all five truncated drafts, which is the behaviour I wanted. That run's outputs were overwritten, and a probe in the final notebook did not reproduce the stop (0 of 3 isolated verbatim-copy prompts were filtered), so the filter appears context-dependent; I report the observation rather than claim it is reproducible. I fixed it by asking agents to paraphrase findings in short sentences and by retrying once on a truncated generation. Paraphrasing lowered word-overlap support to 0.612 while LLM-judged faithfulness stayed at 1.0, which shows why a lexical proxy should not be the only grounding metric. Other honest weaknesses: the orchestrator chose an ineligible agent in about 18% of decisions (usually jumping to the writer early), the guardrails are regular expressions that a paraphrased injection could evade, sixteen cases are too few for confidence intervals, retrieval queries were labelled by me, and each run takes about a minute. With more time I would validate on a licensed clinical cohort with subgroup fields, add an embedding-based injection classifier and parallelize the gathering agents.

Professional Relevance

This project reflects how I would ship AI in a regulated setting: start from a contract, give each agent least privilege, verify outputs deterministically, keep a human approval step and leave a signed record. It shows technical skill (MCP, LangGraph, calibrated machine learning, retrieval), analytical skill (hypothesis tests, calibration and failure analysis), ethical skill (safeguards tied to specific harms) and communication skill (a product UI, buyer documentation and this paper). The live app runs on Streamlit Community Cloud, and the codebase is packaged as a starter kit under the MIT license without the educational-use dataset. My next steps

are external validation, a networked A2A deployment with authentication, and a compliance review before any clinical pilot.

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